

Using Trained Machine Learning Algorithms for Classification of Normal and Abnormal Heart Sounds from Phonocardiogram Data

By: Iesha Phillips

Adaptive Algorithm for Machine Learning

Southern Methodist University

December 14, 2025

Abstract

When analyzing and detecting heart sounds, traditional methods rely on listening with a stethoscope and clinical experience, which can be inconsistent. This project focuses on improving heart sound analysis by applying machine learning techniques to classify normal and abnormal heart sounds using phonocardiogram recordings. Heart sounds contain important information about cardiac function, and the use of automated detection of murmurs can conduct faster screening. Using the PhysioNet CinC 2016 dataset, PCG signals were processed in the time domain, frequency domain, and time frequency spectrogram domain to examine how different signal representations can affect performance of machine learning models. I explored two approaches; an image-based convolutional neural network trained on spectrograms and a one-dimensional convolutional neural network trained on filtered data. The accuracy was evaluated using confusion matrices, sensitivity, specificity, and per valve analysis. The results demonstrate how signal representation and model architecture influence murmur detection and highlights the tradeoffs for heart sound detection.

Introduction

Cardiovascular disease is one of the main leading causes of death worldwide, making early identification of abnormal heart sounds a priority in today's healthcare system. One of the most used techniques to evaluate heart function is called auscultation, which involves listening to bodily noises through a stethoscope. While auscultation is inexpensive and noninvasive, it relies on clinician experience and interpretation. Heart sounds can vary significantly between patients where each murmur can be subtle, meaning environmental noise can cause major interference. Even with trained clinicians, identifying abnormal heart sounds can have difficulties. These limitations have motivated growing interest in automated approaches that use machine learning to assist with heart sound analysis and murmur detection.

Phonocardiograms or PCG recordings provide a digital representation of heart sounds by capturing acoustic vibrations produced during the mechanical activity of the heart. In a normal cardiac cycle, the first heart sound S1 occurs when the mitral and tricuspid valves close at the beginning of systole, while the second heart sound S2 occurs when the aortic and pulmonic valves close at the end of systole. Additional heart sounds can appear depending on the patient's condition, such as S3, which is associated with rapid ventricular filling and can indicate heart failure in adults, and S4, which occurs just before systole and is linked to reduced ventricular compliance. Heart murmurs are made by rapid and choppy blood flow and can be a sign of

diseases or damage to the heart. These features are distributed across both time and frequency domains, making automated methods a vital source for detection.

Due to the complex temporal and spectral patterns present in heart sounds, machine learning methods are well fit for murmur detection. Convolutional neural networks are very effective considering they can learn features directly from data rather than relying on manual engineering of the features. In heart sound analysis, two primary signal representations are commonly used. One dimensional CNNs applied directly to waveform data (*Figures 1 & 2*) preserves temporal detail and avoids transforming the signal into the frequency domain (*Figures 3 & 4*). On the other hand, time-frequency representations such as spectrograms (*Figures 5 & 6*) allow two dimensional convolutional networks to treat heart sounds as images and learn the combination of temporal and spectral features. By comparing the two approaches, I am able to determine how different representation of data affects classification accuracy. This project uses the PhysioNet CinC 2016 heart sound dataset, which contains PCG recordings collected from multiple auscultation sites, including the aortic, pulmonic, tricuspid, and mitral valves, along with patient features such as weight, height, gender, and age.

Each valve corresponds to a distinct region of the heart, and murmurs are presented differently depending on their origin. A central goal of this project was to evaluate model performance not only overall but also on a per valve basis. Per valve analysis measures how accurately a model detects murmurs at each specific auscultation site. This is important considering a model that performs well overall may still struggle independently depending on the valve location. By analyzing sensitivity and specificity separately for the aortic, pulmonic, tricuspid, and mitral valves, the aim was to identify which signal representations and architectures best capture valve specific features. Through these comparisons, this project examines how architectural choices, signal representation, and recording location influence murmur detection accuracy. The goal is not only to build effective classifiers but also to understand which approach strongly impact performance the best and can help support the use of machine learning tools for automated heart sound screening in real world clinical settings in the future.

Literary Survey

The detection of normal and abnormal heart sounds using machine learning has become an increasingly popular area of research in recent years. This interest is driven by the interdisciplinary nature of the problem, which lies at the intersection of biomedical engineering, signal processing, and artificial intelligence. Researchers from across many different fields have explored a wide range of techniques to improve the reliability of heart murmur detection using phonocardiogram recordings.

A clinical example of this research is a study published in the Journal of the American Heart Association, titled, *Deep Learning Algorithms to Detect Murmurs Associated With Structural Heart Disease*, that investigates automated murmur detection using a digital stethoscope platform. In this work, the authors train a convolutional neural network on a large dataset of heart sounds and compare the performance of the model to trained clinicians. For murmur classification, the researchers observed high test performance, with a sensitivity of 85.6% and a specificity of 84.4%. One of the major strengths of this study is its emphasis on real world data and reinforces the idea that machine learning systems could be used as main screening tools in primary healthcare, especially where access to cardiology specialists may be limited. This research primarily focuses on murmur detection as a binary classification task and does not explore how predictions vary across specific valve locations.

In Eran Hadad's Final Project for the MSc in Machine Learning and Data Science, titled *Murmur Detection in Heart Sounds: Evaluating CNN and ResNet Architectures with Spectrogram Analysis*, her work presents a complete machine learning study that involves signal preprocessing, feature extraction, model training, and evaluation. In addition to standard spectrograms, Hadad explores time-frequency representations derived from wavelet transforms and Mel-based features to capture both localized temporal changes and meaningful frequency content in heart sounds. The wavelet transform is used to better represent transient murmur components that may vary across the cardiac cycle, while Mel spectrograms emphasize lower-frequency regions where most heart sound energy is concentrated. These representations are then used as inputs to convolutional neural networks called ResNet-18. The inclusion of ResNet-18 allows the model to learn more complex hierarchical patterns through residual connections that improve feature extraction from the transformed representations. For both methods, she received an accuracy score of above 90%. Hadad's results show that combining wavelet-based and Mel-based time-frequency features with CNN can reliably distinguish between normal and abnormal phonocardiogram recordings. This work demonstrates how alternative feature representations can enhance spectrogram-based learning and provides motivation for comparing different signal representations, which aligns closely with the goals of my own project.

A more signal processing focused approach is highlighted by Alkhodari, Hadjileontiadis, and Khandoker in their IEEE Journal of Biomedical and Health Informatics, titled, *Identification of Congenital Valvular Murmurs in Young Patients Using Deep Learning-Based Attention Transformers and Phonocardiograms*. This study focuses on valvular murmur detection. Using the PhysioNet CinC 2022 CirCor DigiScope dataset, the authors analyze pediatric phonocardiograms recorded at the four auscultation sites (aortic, mitral, pulmonic, and tricuspid valves). Instead of converting signals into spectrogram images,

they apply a discrete wavelet to transform compact features such as energy, variance, standard deviation, and waveform length. These features are stacked across channels and passed into an attention-based transformer model that learns long range temporal relationships within the data. The study reports strong performance across multiple evaluations receiving high accuracy of 89.57%. Through their analysis, the authors demonstrate that different valve channels contribute differently to murmur predictions. This work is especially relevant to my project considering emphasizing the importance of valve specific information and time-frequency representations.

Even though the focus has been on CNN models, machine learning is a very broad field, and it is important to understand how other methods perform before deciding on a final approach. This comparison is shown in the International Journal of Advanced Computer Science and Applications paper by Hussein AlBazar, Mallouhy, and Al-Hebsh, titled, *Deep Learning in Heart Murmur Detection: Analyzing the Potential of FCNN vs. Traditional Machine Learning Models*. In their research, the authors extract frequency-domain and statistical features from phonocardiogram signals and evaluate several traditional machine learning models, including Support Vector Machines, Random Forests, AdaBoost, and Decision Trees. These models performed very well, with accuracies ranging from about 85% for SVMs to 89% for Decision Trees. As a comparison, they also evaluated a fully convolutional neural network, where the accuracy increased to 99%. This result shows that traditional machine learning models can still be beneficial in terms of time and cost efficiency while achieving strong performance. However, the clear accuracy jump from the traditional methods to the FCNN highlights the advantage of deep learning for this problem. This difference supports my decision to focus on CNN approaches in this project, as they are ideal for the complexity of heart sound signals.

Overall, these studies provide valuable insight and illustrate the wide range of strategies that have been explored for automated heart murmur detection, helping to clarify the design choices made in this project. The literature highlights important tradeoffs between model interpretability, signal preprocessing, and clinical relevance. Motivated by these findings, my project draws inspiration from multiple approaches and incorporates elements from each in order to better understand how machine learning models can be improved for clinical use.

Approach

The methodology for this project focused on designing, implementing, and comparing two machine learning models for heart murmur detection using phonocardiogram recordings. The first approach uses time–frequency representations in the form of spectrograms trained using a two-dimensional convolutional neural network, while the second approach trains directly on filtered waveform data using a one-dimensional convolutional neural network. Both approaches distinguished the data in Murmur and No Murmur classes. Comparing these two frameworks allows for a direct evaluation of how signal representation influences murmur detection performance and helps identify which characteristics of heart sounds are most important for classification.

For both approaches, preprocessing began with applying a bandpass filter between 20 Hz and 800 Hz. This frequency range was chosen because the main energy of normal heart sounds (S1 and S2) as well as most murmurs, lies within this range. A Butterworth filter was implemented due to its maximally flat passband response, which minimizes distortion and helps preserve murmur characteristics. Zero-phase filtering was applied to avoid introducing phase shifts that could distort the timing between heart sounds. This is important when distinguishing systolic and diastolic structures. To add on, class imbalance was also addressed during preprocessing, since recordings without murmurs significantly outnumber murmur recordings. To reduce bias, the training set was balanced by keeping all Murmur samples and down sampling the No Murmur class. This decision helps improve sensitivity to murmurs while still maintaining an appropriate set of normal heard sounds. The validation and test sets were left unbalanced to better reflect real-world screenings.

For the one-dimensional time-domain model, additional preprocessing steps were used to standardize temporal characteristics across recordings. All signals were resampled to a sampling rate of 2000 Hz. This provides sufficient temporal resolution while keeping computational cost low. Each waveform was then trimmed or zero-padded to a fixed length of 4000 samples. The duration was chosen considering it captures multiple cardiac cycles for most heart rates, allowing the model to learn repeated systolic and diastolic patterns rather than in different cardiac events. A fixed input length is also required for stable training of convolutional neural networks that include pooling layers. After length standardization, each waveform was z-normalized by subtracting its mean and scaling its maximum absolute value. This normalization step reduces variance caused by external factors such as background noise.

For the spectrogram approach, parameters for the Short-Time Fourier Transform (STFT) were chosen to balance time and frequency resolution in a way that aligns with cardiac acoustics. The spectrogram magnitudes were converted to a logarithmic dB scale so that large amplitude differences were compressed, allowing murmur-related spectral patterns to become clearer. The spectrograms were resized to 224×224 pixels to match common input

dimensions used in convolutional neural networks originally developed for image classification. They were then stored as grayscale images and later converted to three-channel format during training to maintain compatibility with standard image input layers.

The CNN architectures for both approaches were designed to learn meaningful patterns in the data without overfitting and maintaining a balance between model complexity and generalization. The spectrogram-based CNN follows common image classification norms and uses three convolutional layers with increasing filter depth to progressively capture low-level spectral edges, mid-level frequency patterns, and higher-level murmur structures. Each layer is followed by batch normalization, ReLU activation, and max pooling, which reduces spatial resolution while preserving the most important features. Dropout layers were introduced in the deeper layers to limit overfitting, and global average pooling was used in place of large fully connected layers to reduce the number of trainable parameters and improve robustness. On the other hand, the one-dimensional CNN focuses on temporal structure by using wider convolutional kernels that better capture the duration and spacing of heart sounds and murmurs. Pooling layers gradually reduce temporal resolution while maintaining essential timing information, and dropout is also used to improve model stability. Both models were trained using the adam optimizer, with learning rates of 1×10^{-3} for the spectrogram CNN and 3×10^{-4} for the time domain CNN, batch sizes of 32 and 64, and early stopping based on validation performance. The systems were evaluated using confusion matrices and per valve analysis by auscultation sites (aortic, pulmonic, tricuspid, and mitral), measuring sensitivity, and specificity.

Compared to prior work, this project differs by directly comparing time-domain and time-frequency CNN approaches using the same dataset, preprocessing steps, and evaluation metrics. Many studies focus on a single signal representation, which limits understanding of how different representations impact model performance. In this work, the one-dimensional model offers the advantage of lower computational cost and preserves raw temporal information that reflects the natural structure of heart sounds. While the spectrogram-based approach requires additional preprocessing and higher computational cost, it provides a more in depth representation that captures both temporal and spectral characteristics of normal and abnormal heart sounds. Overall, this approach provides a clearer understanding of how signal representation influences murmur detection performance.

Results

One-Dimensional Time-Domain CNN

The one-dimensional convolutional neural network was evaluated to assess how well raw filtered phonocardiogram waveforms support automated murmur detection when only time-domain information is used. After balancing the training data, the final training set consisted of 125 Murmur recordings and 100 No Murmur recordings. The model was trained using the adam optimizer for up to 60 epochs, and training converged without early stopping. Validation accuracy stabilized at approximately 61.07% (*Figure 7*), indicating moderate generalization to unseen data and suggesting that the model was able to learn meaningful temporal patterns although there were still clear limitations.

On the testing set, the 1D CNN achieved an overall sensitivity of 0.704 and a specificity of 0.667. This means the model correctly identified 70.4% of murmur cases while correctly classifying 66.7% of normal heart sounds with no murmurs. The confusion matrix (*Figure 8*) shows that while the model was reasonably effective at detecting murmurs, it also produced a noticeable number of false positives. This emphasizes the difficulty of distinguishing subtle variations within the waveforms that are associated with murmurs from actual normal heart sounds.

Per-valve analysis revealed clear variability in performance across auscultation sites (*Table 1*). At the aortic valve (AV), which contained the largest number of test recordings of 113, the model achieved a sensitivity of 0.696 and a specificity of 0.667. At the pulmonic valve (PV), sensitivity dropped to 0.500 with a specificity of 0.714 across 16 recordings. Results for the mitral (MV) and tricuspid (TV) valves showed extreme sensitivity values of 1.000 and specificities of 0.000. However, these results were based on only two MV recordings and one TV recording, which means they are not as meaningful as the previous two sites. Overall, this indicates that the AV was the most reliable site for the model due to the high number of recordings and its featural content.

Two-Dimensional Image Processing Spectrogram CNN

The spectrogram-based convolutional neural network was evaluated to determine whether incorporating time–frequency information improves murmur detection performance. After balancing the training data, the final training set consisted of 882 Murmur and 882 No Murmur spectrogram samples. The network was trained using the adam optimizer with a learning rate of 1×10^{-3} and a batch size of 32 for 40 epochs, with L2 regularization applied to improve generalization. Validation accuracy stabilized at approximately 78.29% (*Figure 9*),

For the testing set, the spectrogram CNN achieved an overall sensitivity of 0.711 and a specificity of 0.798. This indicates that 71.1% of murmur cases were correctly detected while 79.8% of normal recordings were accurately classified. The confusion matrix (*Figure 10*) shows a clear reduction in false positives, demonstrating that the model was better able to distinguish normal heart sounds from murmurs when frequency information was included.

Per-valve analysis demonstrated stable and consistent performance across all major auscultation sites (*Table 1*). The highest valve performance in this project was the the mitral valve (MV) site, with a sensitivity of 0.769 and a specificity of 0.857 across 234 recordings. At the aortic valve (AV), the model has similar statistics, a sensitivity of 0.739, and a specificity of 0.783 across 230 test recordings. At the pulmonic valve (PV), the model achieved a sensitivity of 0.700 and a specificity of 0.721 across 212 recordings, The tricuspid valve (TV) showed the lowest sensitivity at 0.619 but maintained a high specificity of 0.833 across 198 samples. Overall, since the performance statistics were relatively close across valves, all auscultation sites appear to be important for the two-dimensional CNN, with MV providing the most informative signals for murmur detection.

Model Comparison

Both models were able to detect heart murmurs effectively; however, each showed clear differences in consistency and clinical reliability. *Table 2* compares the detection accuracy of the two approaches. In the time-domain waveform model, class balancing introduced a noticeable tradeoff where increasing murmur detection led to a decrease in No Murmur accuracy, resulting in more false positives. This tradeoff was not as noticeable in the spectrogram-based model since it maintained a higher specificity; however, the sensitivity still increased as desired. One reason for this difference is that the spectrogram CNN was trained on a significantly larger dataset. This allowed the model to learn more complex patterns within the heart sound. On the other hand, the one-dimensional model was trained on a relatively small number of recordings, which limited its ability to generalize across valve locations. As a result, the per valve analysis for the time-domain model showed that the aortic valve (AV) site was the most informative while the others were affected by the limited sample size. The spectrogram CNN demonstrated more consistent performance across valves, mostly at the aortic and mitral sites. Overall, these results suggest that both models are useful and can incorporate some improvements. While time-domain models can capture useful temporal information, incorporating frequency content leads to more reliable murmur detection, making time–frequency approaches better for this problem.

Conclusion

In this project, I explored how different signal representations and convolutional neural network architectures affect automated heart murmur detection using phonocardiogram recordings. I did this by directly comparing a one-dimensional time domain CNN and a two-dimensional spectrogram CNN while keeping the preprocessing, training, and evaluation consistent between the two. Through this work, I learned how important heart sound information is spread across both time and frequency, and how the choice of representation of the signal can strongly influence how well a model performs. The results showed that while both approaches are similar in statistic, the time-domain model was able to learn temporal patterns related to systole and diastole but struggled with specificity and consistency across valve locations. While the spectrogram model produced more stable and clinically reliable results by capturing both temporal and spectral features of murmurs. Working on this project strengthened my understanding of signal processing and machine learning for biomedical use. I gain skills in filtering, normalization, and time-frequency analysis, as well as overcoming challenges such as class imbalance, regularization, and model evaluation. Future work could build on this by increasing the dataset size, exploring multichannel recordings, or testing more in-depth CNN models using transformers or wavelets. While higher time complexity models currently perform best, this project also highlights the value of continuing to improve low cost one dimension time domain and traditional approaches to solve the problem of a more efficient way to detect murmurs to save the lives of many around the world.

Figures

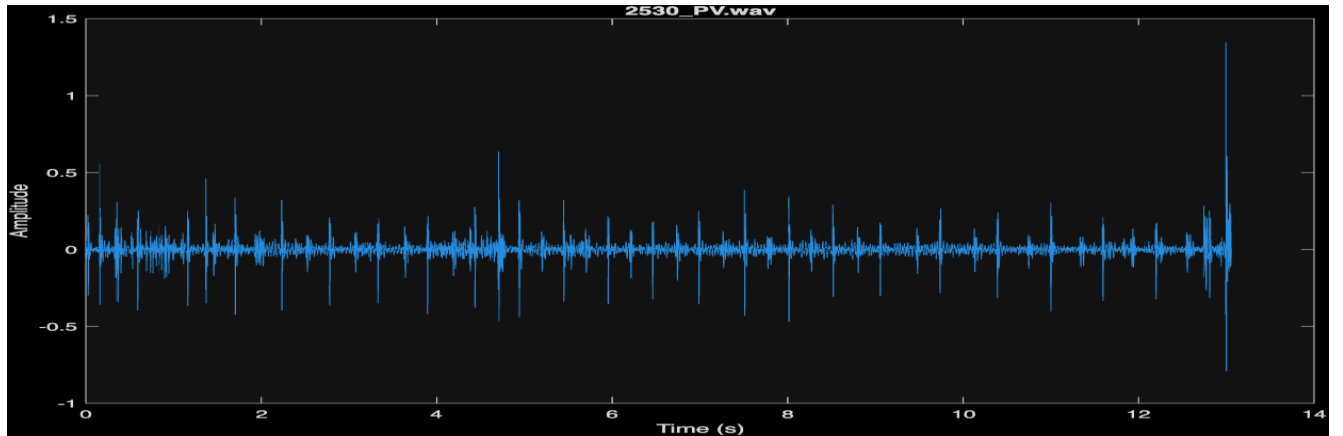


Figure 1: Time-domain of patient *without* murmur

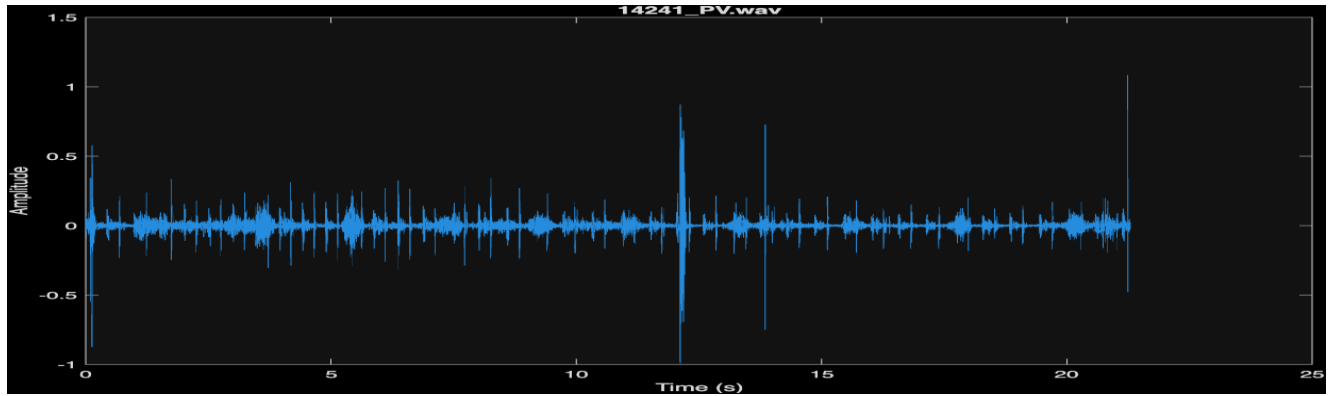


Figure 2: Time-domain of patient *with* murmur.

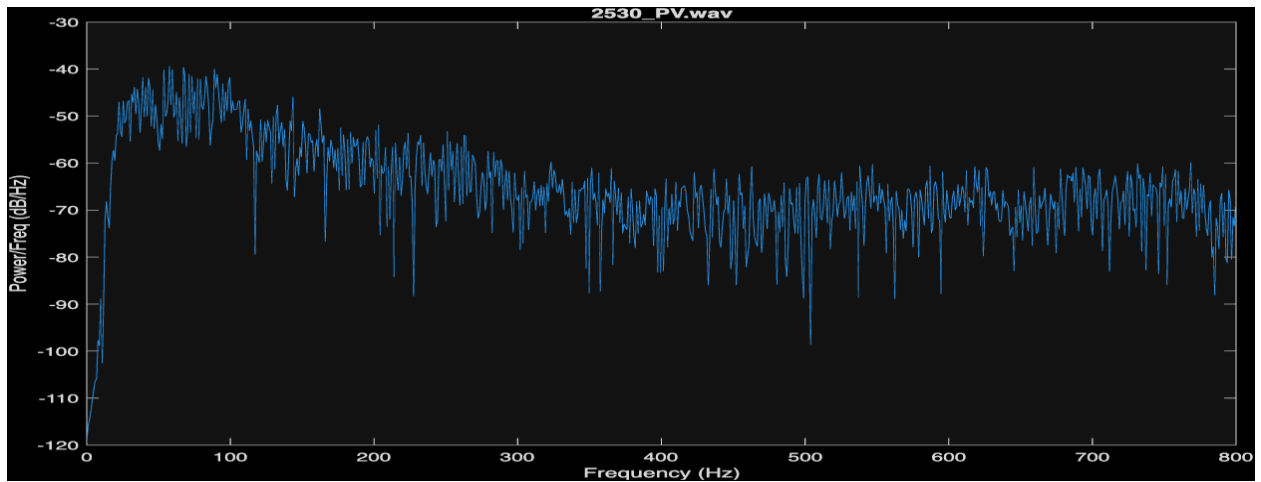


Figure 3: Frequency domain of patient *without* murmur

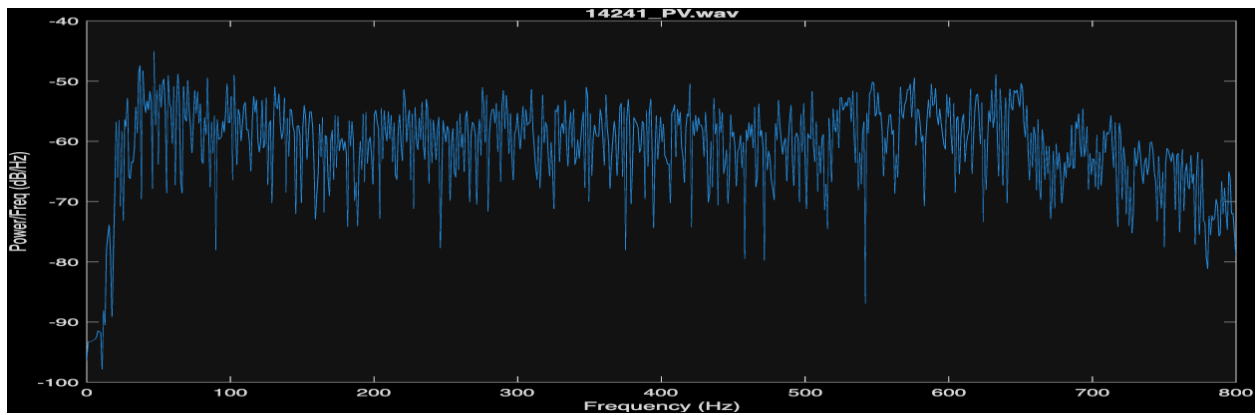


Figure 4: Frequency domain of patient *with* murmur

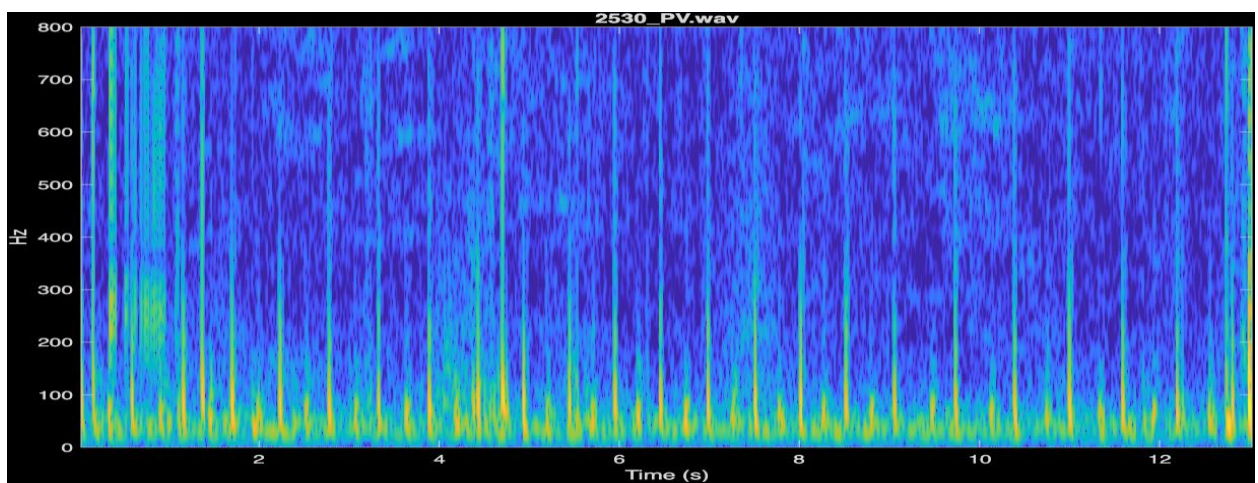


Figure 5: Time-Frequency domain of patient *without* murmur

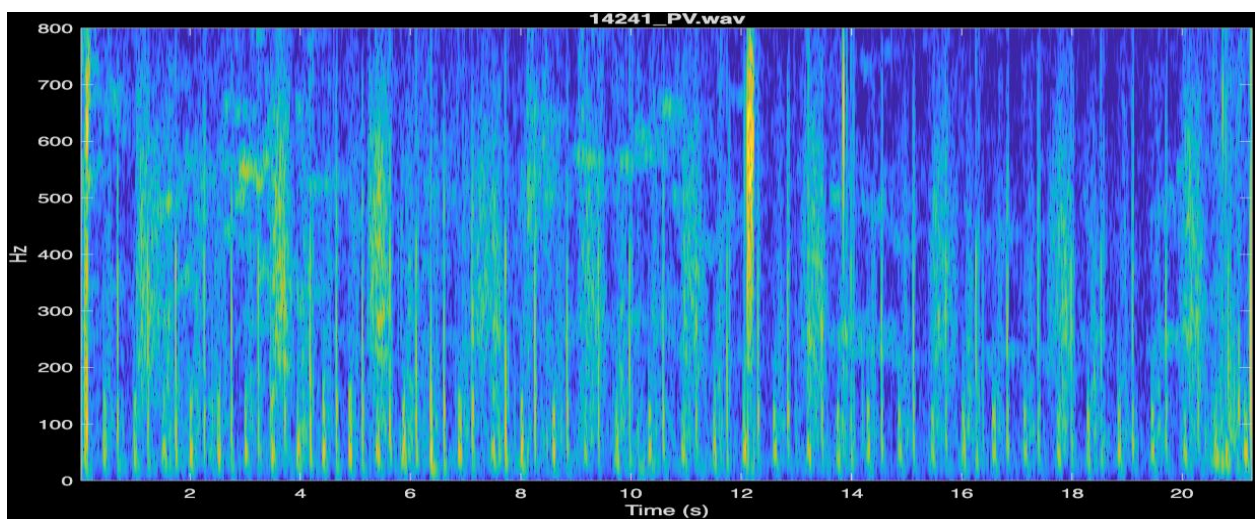


Figure 5: Time-Frequency domain of patient *with* murmur

Results

One-Dimensional Time-Domain CNN Figures

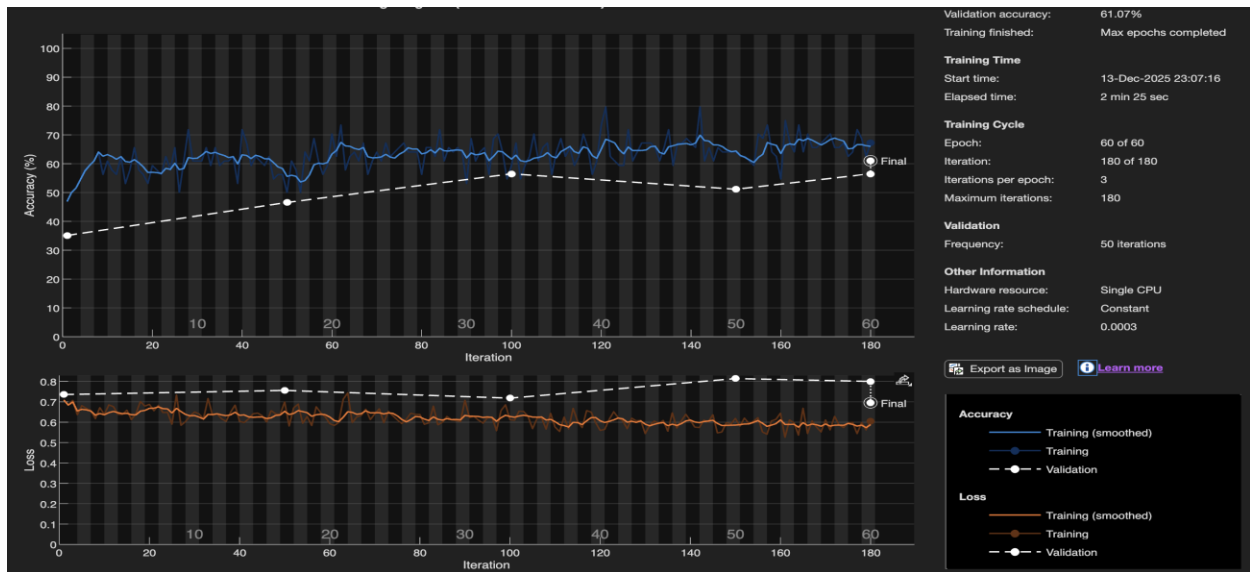


Figure 7: Training and validation accuracy loss curves for the One-Dimensional Time-Domain Convolution Neural Network

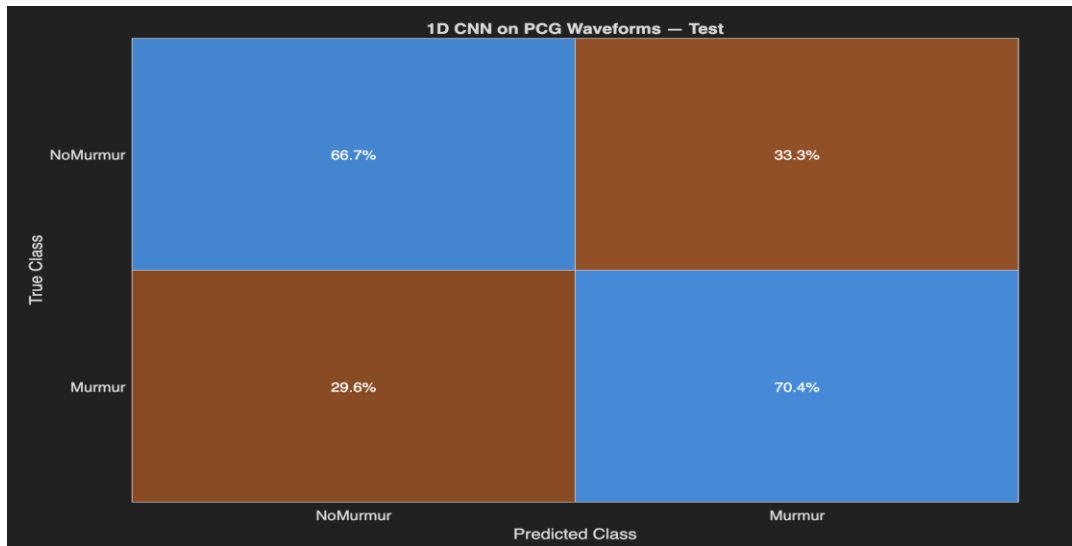


Figure 8: One-Dimensional Convolutional Neural Network on PCG Waveforms Confusion Matrix

Two-Dimensional Image Processing Spectrogram CNN Figures

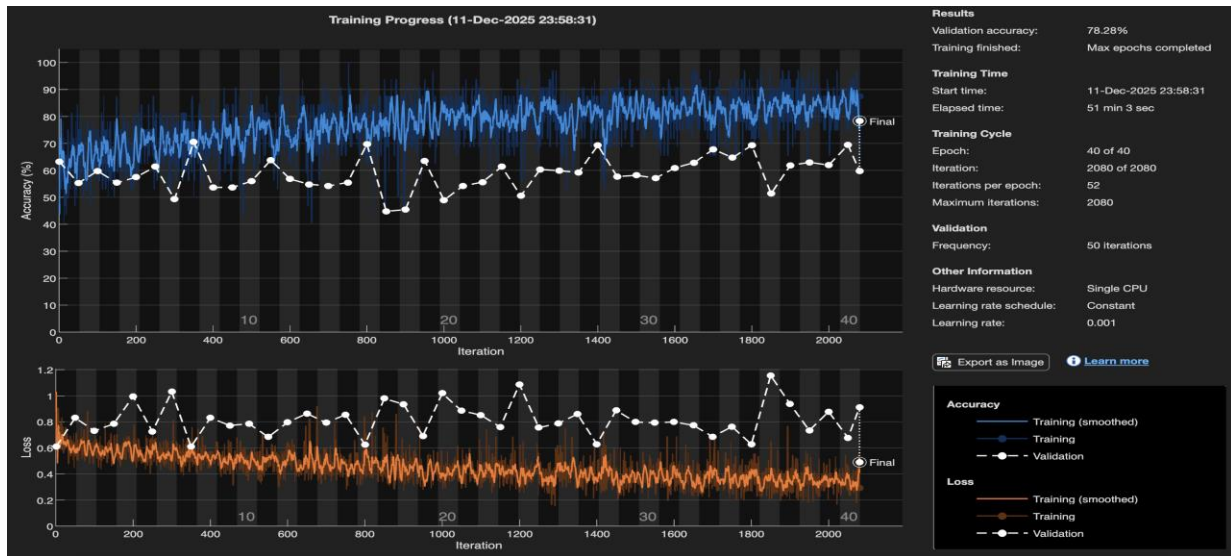


Figure 9: Training and validation accuracy loss curves for the 2D spectrogram Convolutional Neural Network

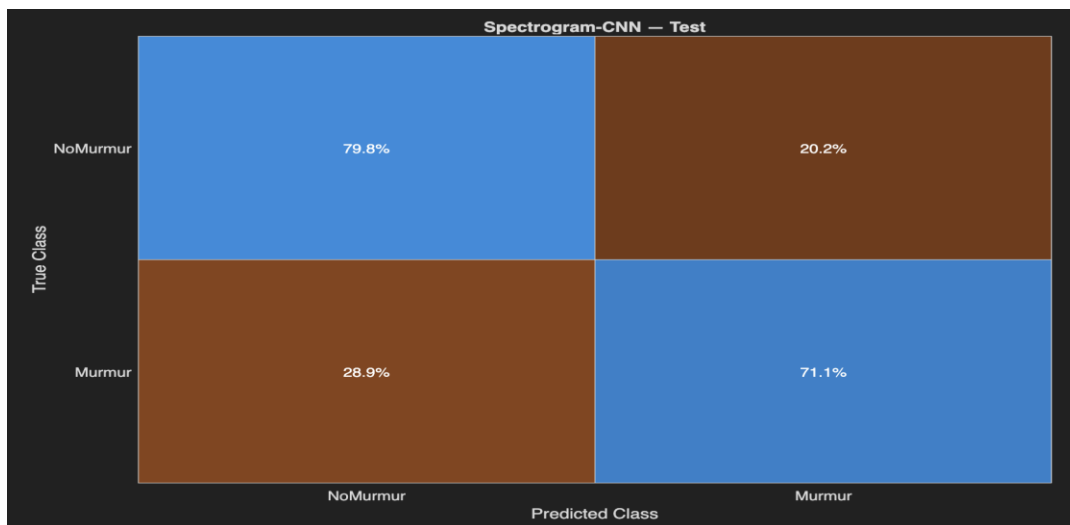


Figure 10: Two-Dimensional Spectrogram Convolution Neural Network Confusion Matrix

Valve Sites	Model	Murmurs Detected (%)	No-Murmurs Detected (%)
Aortic (AV)	1D Time-Domain	69.60%	66.70%
	2D Spectrogram	73.90%	78.30%
Mitral (MV)	1D Time-Domain		
	2D Spectrogram	76.90%	85.70%
Pulmonic (PV)	1D Time-Domain	50.00%	71.40%
	2D Spectrogram	70.00%	72.10%
Tricuspid (TV)	1D time-Domain		
	2D Spectrogram	61.90%	83.30%

Table 1: Per Valve Analysis Model Comparison

Model	Murmurs Detected (%)	No-Murmurs Detected (%)
1D Time-Domain CNN	70.40%	66.70%
2D Spectrogram CNN	71.10%	79.80%

Table 2: Overall Model Comparison

References

Classification of Heart Sound Recordings: The PhysioNet/Computing in Cardiology Challenge 2016 <https://physionet.org/content/challenge-2016/1.0.0/training-a/#files-panel>

Chen W, Zhou Z, Bao J, Wang C, Chen H, Xu C, Xie G, Shen H, Wu H. Classifying Heart-Sound Signals Based on CNN Trained on MelSpectrum and Log-MelSpectrum Features. *Bioengineering (Basel)*. 2023 May 25;10(6):645. doi: 10.3390/bioengineering10060645. PMID: 37370576; PMCID: PMC10294824. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10294824/>

Harvey, W. P. (1964). Heart sounds and murmurs. *Circulation*, 30(2), 262–271. <https://doi.org/10.1161/01.CIR.30.2.262>

M. Alkhodari, L. J. Hadjileontiadis and A. H. Khandoker, "Identification of Congenital Valvular Murmurs in Young Patients Using Deep Learning-Based Attention Transformers and Phonocardiograms," in *IEEE Journal of Biomedical and Health Informatics*, vol. 28, no. 4, pp. 1803-1814, April 2024, doi: 10.1109/JBHI.2024.3357506. <https://ieeexplore.ieee.org/document/10412339?denied=>

Wu, T., Huang, Z., Li, S., Zhao, Q., & Pan, F. (2024). Heart Murmur Quality Detection Using Deep Neural Networks with Attention Mechanism. *Applied Sciences*, 14(15), 6825. <https://doi.org/10.3390/app14156825>

Hadad, E. (2025). Murmur detection in heart sounds: Evaluating CNN and ResNet architectures with spectrogram analysis (Master's final project). MSc in Machine Learning and Data Science. https://cs-msc-mllds.org/wp-content/uploads/2025/09/MLDS_Final_Project_Heart_Sounds_Jan_2025.pdf

AlBazar, H., Mallouhy, R., & Al-Hebsh, A. (2025). Deep learning in heart murmur detection: Analyzing the potential of FCNN vs. traditional machine learning models. *International Journal of Advanced Computer Science and Applications*, 16(2). https://thesai.org/Downloads/Volume16No2/Paper_128-Deep_Learning_in_Heart_Murmur_Detection.pdf

Prince, J., Maidens, J., Kieu, S., Currie, C., et al. (2023). Deep learning algorithms to detect murmurs associated with structural heart disease. *Journal of the American Heart Association*, 12(20), e030377. <https://doi.org/10.1161/JAHA.123.030377>

Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). ImageNet classification with deep convolutional neural networks. In *Advances in Neural Information Processing Systems 25 (NIPS 2012)* (pp. 1097–1105). https://proceedings.neurips.cc/paper_files/paper/2012/file/c399862d3b9d6b76c8436e924a68c45b-Paper.pdf

Jiang Z, Song W, Yan Y, Li A, Shen Y, Lu S, Lv T, Li X, Li T, Zhang X, Wang X, Qi Y, Hua W, Tang M, Liu T. Automated valvular heart disease detection using heart sound with a deep learning algorithm. Int J Cardiol Heart Vasc. 2024 Mar 5;51:101368. doi: 10.1016/j.ijcha.2024.101368. PMID: 38482387; PMCID: PMC10933456. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10933456/>

Phillips, I. P., Hoang, C., & Vieira, G. B. (2025). Using a trained machine learning algorithm for microparticle tracking on an inhomogeneous disk array background.

